Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

14.33 Research Paper — Final Presentation

Nicholas Liu

MIT Department of Economics

Spring 2026

Motivation & Research Question

The puzzle

- Epic Games Store (EGS) has given away 541 games for free since 2018 — its primary differentiation lever against Steam
- Standard theory predicts a zero-price substitute on a rival platform diverts players *away* from Steam
- But in January 2026, New Blood CEO Dave Oshry reported that giving away *Blood West* on EGS caused Steam sales to surge $\sim$200% the same day

Research question

Does an EGS free giveaway *increase* or *decrease* Steam player engagement for the same game — and how does this evolve over time?

Two competing mechanisms

Advertising: Giveaway generates media buzz $\rightarrow$ players flock to Steam (primary discovery/community layer) $\rightarrow$ positive short-run spillover

Substitution: Free copy on Epic gradually diverts playtime away from Steam $\rightarrow$ negative medium-run effect

Both may operate, but at different horizons.

Data

EGS Giveaway History

- Kaggle dataset (Dongre, 2025): every free game on EGS Dec 2018–2025
- 541 events $\rightarrow$ 471 unique titles; 70 repeat appearances
- 63 titles given away more than once (30 within 12 months = "proximate")

Steam Player Data

- Monthly average concurrent players scraped from SteamCharts for each matched game
- Panel: July 2012 – February 2026
- Outcome: $\ln(\texttt{avg_players} + 1)$

Final Panel

Games in final panel	446
Game-month observations	38,642
Median game: months of data	88
Mean avg. players/month	1,584
(median)	65
Mean $\ln(\texttt{avg_players} + 1)$	4.43

Player counts are heavily right-skewed — 101 games have pre-treatment median $<$ 10 players. Log transformation compresses the tail.

Matching: RapidFuzz fuzzy string matching (WRatio $\geq$ 88) + manual review; 8 titles confirmed absent from Steam

Identification Strategy

Design: Staggered DiD Event Study

- Treatment = game i's first EGS giveaway, staggered across 80 calendar months (Dec 2018 – late 2025)
- **Control group:** not-yet-treated games — games awaiting future giveaways serve as controls for currently treated games
- No external never-treated group needed (but this is the main limitation)

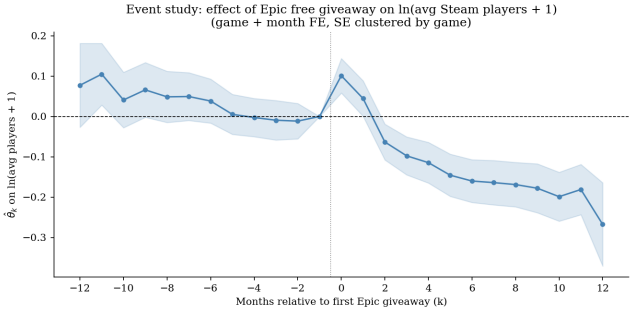
Key assumption (parallel trends)
In the absence of treatment, log player trajectories of different cohorts would have trended identically after conditioning on game & calendar-month FE. Partially testable via pre-period coefficients.

Four Estimators (for robustness)

1. **TWFE** — primary specification; game + calendar-month FE; event window $k \in [-12, 12]$
2. **LP-DiD** (Dubé et al., 2023) — local projections using only not-yet-treated controls at each k; robust to heterogeneous effects
3. **Stacked DiD** (Cengiz et al., 2019) — cohort-specific balanced panels; avoids negative-weight bias
4. **CS-ATT** (Callaway & Sant'Anna, 2021) — cohort-specific ATTs; most conservative

Randomization placebo: 200 permutations of treatment dates to validate the $k = 0$ spike

Results: Two-Phase Dynamic



Pooled TWFE. Game + calendar-month FE. $N = 38{,}641$; 446 games. Reference: $k = -1$.

Phase 1 — Advertising spike

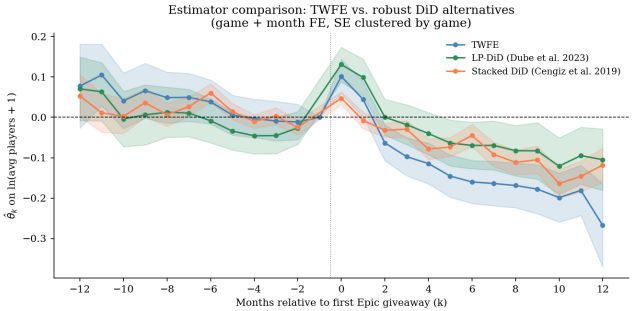
- $k = 0$: $\hat{\theta}_0 = +0.101$ ($p < 0.001$) $\Rightarrow$ $\approx$10.6% more Steam players
- Consistent across all estimators: +5 to +14% at $k = 0$
- Placebo $p < 0.001$; actual estimate is 2.7× the 95th percentile of null

Phase 2 — Substitution reversal

- Turns negative by $k = 2$ (−6.3%, $p = 0.005$)
- −10 to −23% below baseline by $k = 12$ across estimators
- Significant at 5% under all three robust estimators from $k = 5$ onward

Repeat giveaways: $\hat{\delta} = 0.0001$ ($p = 0.999$)

Robustness: All Four Estimators Agree



TWFE, LP-DiD (Dubé et al., 2023), Stacked DiD (Cengiz et al., 2019). Reference: $k = -1$.

k	TWFE	LP-DiD	Stacked
0	$+0.101^{***}$	$+0.131^{***}$	$+0.047^{***}$
1	$+0.045^{*}$	$+0.099^{***}$	-0.008
2	-0.063^{**}	0.000	-0.032^{***}
5	-0.145^{***}	-0.064^{*}	-0.073^{***}
10	-0.199^{***}	-0.121^{***}	-0.164^{***}
12	-0.267^{***}	-0.105^{**}	-0.118^{***}

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *SE clustered by game.*

All estimators agree: short-run positive spike, sustained medium-run reversal. Spread at $k = 0$ (0.047 to 0.131) bounds the true ATT.

Takeaways, Caveats & Future Work

Main takeaway

Both mechanisms are correct — at different horizons:

- **Short run:** Giveaway announcement → media coverage → Steam awareness spike (+5–14%)
- **Medium run:** Free Epic copy gradually substitutes Steam engagement (−10–23% by month 12)
- Epic's free-game program may inadvertently function as Steam advertising

Implications

- For developers: net benefit depends on Epic payment vs. long-run Steam erosion
- Repeat giveaways add no marginal value ($\hat{\delta} \approx 0$)

Caveats

- **No never-treated group** — identification rests entirely on timing variation; parallel trends untestable in absolute sense
- **Player counts $\neq$ purchases or revenue** — SteamCharts measures concurrent players, not sales
- **Genre heterogeneity** — single-player vs. multiplayer titles have different substitution dynamics (pooled estimate conflates them)
- **DLC/bundle events** — 15 giveaways were DLC mapped to base-game Steam IDs

Future work

- Genre-stratified analysis (single-player vs. multiplayer)
- Study effects on Steam *sales* volume